

SRI LAKSHMI PARVATHI ALLU

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SUMMARY

Data Engineer and architect with robust SQL proficiency, specializing in scalable data pipelines and deep data analysis from diverse sources, including legacy systems and cloud storage. Proven record of enhancing operational efficiency and bridging business with engineering teams in finance environments. Adept at gathering requirements and articulating innovative, process-improving solutions.

TECHNICAL SKILLS

- **Data Analysis & Solutions:** Data analysis, Solution articulation, Articulating solutions, Holistic view of solutions, Decision articulation, Big data
- **Project Coordination & Stakeholder Management:** Process improvement, Requirement gathering, Coordination with teams, Handling multiple requests

WORK EXPERIENCE

Baylor Scott & White Health

Feb 2024 - Present

Data Engineer

Washington

- Optimized ETL workflows with AWS Glue and Apache NiFi to automate data extraction, transformation, and loading processes, enhancing process improvement.
- Improved performance by optimizing SQL queries and data processing workflows for faster analytics.
- Used Tableau to build interactive dashboards for real-time analytics, empowering stakeholders with actionable insights.
- Leveraged AWS services such as S3, Lambda, and Athena to process and analyze patient data in real-time.
- Designed and implemented scalable data pipelines for healthcare data processing, streamlining data flow from source systems to the data warehouse.
- Integrated data from various sources such as EHR systems, clinical databases, and external APIs to create a unified view of patient information.
- Utilized Apache Spark and PySpark for distributed data processing to handle large-scale healthcare datasets.
- Developed and maintained real-time streaming solutions using Apache Kafka to monitor patient data.
- Collaborated with the data science team to integrate machine learning models, enhancing the accuracy of patient readmission predictions.
- Designed and implemented data models for healthcare analytics, improving insights into patient care outcomes.
- Implemented data encryption and masking techniques to ensure HIPAA compliance and enhance data security.
- Utilized Google Cloud Platform (BigQuery, Pub/Sub) for large-scale data storage and processing, improving data accessibility and efficiency.
- Developed data pipelines for ingesting patient data from various medical devices, ensuring seamless integration.
- Built custom APIs to pull data from external sources such as pharmacies and insurance providers, enriching data availability.
- Built data lakes to store unstructured health data, enabling advanced analytics for research teams.
- Set up automated alerts using Prometheus and Grafana to monitor data pipeline performance.
- Ensured data quality and consistency by implementing regular validation checks within the pipeline.
- Assisted in troubleshooting pipeline failures and implemented corrective actions to improve reliability.

Pacific Western Bank

May 2021 - Jan 2023

Data Engineer

CA

- Worked closely with business analysts to gather requirements and deliver actionable insights through custom reports, facilitating effective stakeholder communication.
- Utilized SQL to optimize data extraction and loading from transactional databases, resulting in faster query performance and reduced data processing time.
- Developed ETL pipelines for processing financial transaction data and integrating it into the bank's data warehouse, improving data accessibility for analytics teams.
- Leveraged real-time data streaming technologies such as Apache Kafka and AWS Kinesis to synchronize transactional data across departments.
- Automated data extraction from various banking systems using AWS Glue and Apache NiFi, reducing manual processing time and increasing data accuracy.
- Designed data models for tracking financial transactions, customer behavior, and other key metrics to support strategic decision-making.
- Built data lakes and managed the integration of structured and unstructured financial data, enhancing analytical capabilities for financial reporting.
- Created interactive dashboards using Power BI to monitor daily banking operations and KPIs.
- Developed automated reporting systems to track critical financial data, reducing manual effort for bank analysts.
- Integrated third-party APIs for external financial data enrichment, bolstering internal reporting robustness.
- Conducted performance tuning of SQL queries to ensure rapid data retrieval and reporting.
- Managed version control of data models and scripts using Git and GitHub, ensuring consistent and reliable updates.

- Implemented CI/CD pipelines using Jenkins and GitLab for automated deployment of data engineering solutions, reducing deployment time and errors.
- Managed data security by enforcing data encryption and masking to comply with financial regulations and protect sensitive information.
- Performed root-cause analysis of data pipeline issues and optimized workflows to minimize downtime and improve system reliability.
- Built AWS Redshift clusters to enhance data warehousing and analytical processing capabilities.
- Developed an automated system for transaction fraud detection using machine learning models integrated into data pipelines.
- Ensured data consistency by implementing robust quality checks throughout the data pipeline.
- Collaborated with IT teams to deploy and scale data processing infrastructure using Docker and Kubernetes.
- Provided training and support to junior team members on best practices in data engineering.

FedEx Express

Jan 2018 - May 2021

Jr Data Engineer

Memphis

- Optimized SQL queries and data retrieval workflows to support quick reporting and analytics on large-scale logistics data.
- Built and maintained ETL pipelines for processing logistics and shipping data from various systems, resulting in improved operational efficiency and data accuracy.
- Utilized AWS services such as S3 and Lambda for scalable storage and processing of shipping data.
- Employed Apache Flink for real-time stream processing of parcel delivery statuses.
- Integrated external data sources and APIs to enrich logistics datasets for enhanced operational insights.
- Designed data models for shipping analytics to improve decision-making and operational efficiency.
- Created dynamic dashboards using Tableau and Looker for real-time monitoring of logistics performance.
- Developed automated reporting systems that provided real-time insights into shipping volumes, delays, and customer satisfaction.
- Ensured data quality through rigorous validation checks and automated cleansing of large datasets.
- Deployed containerized applications using Docker to streamline development and deployment processes.
- Utilized Apache Kafka for real-time monitoring of parcel statuses across multiple shipping channels.
- Collaborated with the operations team to develop solutions aimed at minimizing package delivery delays.
- Built a data warehouse using Redshift for comprehensive reporting on global logistics data.
- Automated routine tasks with Cron Jobs to schedule periodic data extractions and transformations.
- Enhanced performance by tuning data pipelines and ensuring efficient resource utilization.
- Participated in cross-functional teams to define data requirements and optimize the data infrastructure.
- Designed cloud-based data architectures for scalable, efficient, and secure logistics data processing.
- Improved data security by implementing encryption and access controls on sensitive shipping data.
- Coordinated with IT and security teams to ensure GDPR compliance in data handling and processing.

Education

Texas Tech University

Masters, Computer Science

Projects

Migration to cloud-Based Data Platform

Feb 2024 - Oct 2024

Baylor Scott & White Health

- The existing on-premises data infrastructure was fragmented, leading to inefficiencies in data access and reporting across departments.
- Led the migration of legacy data systems to a unified cloud-based platform using Snowflake and DBT. This involved designing and implementing ETL pipelines, ensuring data integrity, and optimizing performance.
- Achieved a 40% improvement in data processing speeds and reduced data retrieval times by 50%.
- Enhanced decision-making capabilities through faster and more reliable data access, supporting better patient care and operational efficiency.

Cloud Data Lakehouse Implementation for Financial Analytics

Jun 2021 - Nov 2022

Pacific Western Bank

- The bank's legacy on-premises data systems were siloed, resulting in delayed reporting, regulatory compliance risks, and rising infrastructure maintenance costs.
- Led the design and development of a cloud-native data lakehouse architecture on AWS integrating S3, Glue, Redshift, and Athena. Built automated ETL pipelines using Python, SQL, and Apache Airflow to ingest, clean, and unify financial transaction, customer, and compliance data.
- Reduced data processing and report generation time by 60%, improved data accuracy for internal audits, and enabled self-service analytics across finance and compliance teams.
- Lowered infrastructure and operational costs by approximately \$500K annually through cloud migration, improved regulatory compliance, and enhanced decision-making speed — directly supporting strategic growth initiatives.

COVID-19 Vaccine Logistics Data Pipeline

Apr 2020 - Mar 2021

FedEx Express

- Urgent need for accurate, real-time tracking of PPE and vaccine shipments during the COVID-19 crisis, while maintaining HIPAA compliance.

- Developed emergency ETL pipelines using AWS Glue and Python to prioritize and monitor critical healthcare shipments, with strict data privacy measures.
- Enabled 99% on-time delivery rate for critical shipments during the pandemic response.
- Strengthened FedEx's healthcare logistics reputation, leading to multi-million dollar government contracts for vaccine distribution.

Real-Time Package Tracking System Enhancement

Jan 2018 - Dec 2019

FedEx Express

- Tracking delays and inaccurate shipment updates were impacting customer satisfaction and operational efficiency.
- Built real-time data pipelines using Apache Kafka and AWS Redshift to stream and store live package tracking data from hubs and vehicles.
- Improved tracking update speed by 35% and reduced customer support inquiries by 20%.
- Increased customer satisfaction scores and lowered operational costs associated with manual tracking issue resolutions.

Certifications

- **Microsoft Certifies Fabric Data Engineer Associate:** Issued by Microsoft—April 2025
 - Demonstrated proficiency in building end-to-end data solutions using Microsoft Fabric, including Lakehouse architecture, Delta Lake, Data Factory pipelines, Power BI, and Spark-based transformations.
 - Skilled in integrating diverse data sources, optimizing performance, and applying governance and security best practices within a unified analytics environment.
- **AWS Certified Solution Architect-Associate:** Issued by: Amazon Web Services (AWS)—April 2025
 - Validated expertise in designing distribution, scalable, and secure cloud architecture using core AWS services including EC2, S3, RDS, VPC and IAM.
 - Skilled in architecting fault-tolerant applications, implementing cost-optimized infrastructure, and leveraging best practices for high availability, security, and performance within the AWS Well-Architected Framework.
- **AWS APAC-Solution Architecture Job Simulation:** Issued By: Forage—March 2025
 - Completed a hands-on simulation of real-world AWS Solutions Architecture tasks, including cloud infrastructure design, architecture optimization, and customer use case analysis.
 - Demonstrated skills in cloud migration planning, AWS best practices, and delivering scalable, cost-efficient, and secure cloud solutions aligned with customer requirements across the APAC region.
- **Risk Management – Goldman Sachs:** Issued By: Forage—March 2025
 - Completed a virtual experience simulating real-world risk management tasks at Goldman Sachs, including identifying, assessing, and mitigating financial and operational risks.
 - Demonstrated skills in risk analysis, strategic thinking, and compliance evaluation through case-based activities.
 - Applied industry-standard risk frameworks to assess scenarios, propose data-driven solutions, and support decision-making in high-stakes in financial environments.

Languages

- English (Fluent), Hindi (Fluent), Telugu (Native), German (Intermediate), Urdu (Beginner), Spanish (Beginner)